

Discrete Math — Monday Test

Practice Problems (Rosen, 7th ed.)

Full question text, pulled directly from your textbook and organized by study day. Solutions are in the companion PDF.

Tuesday — §1.1 Propositional Logic & §1.3 Equivalences

§1.1 Propositional Logic

1. What is the negation of each of these propositions?

- (a) Mei has an MP3 player.
- (b) There is no pollution in New Jersey.
- (c) $2 + 1 = 3$.
- (d) The summer in Maine is hot and sunny.

2. Let p and q be the propositions

p : It is below freezing. q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- (a) It is below freezing and snowing.
- (b) It is below freezing but not snowing.
- (c) It is not below freezing and it is not snowing.
- (d) It is either snowing or below freezing (or both).
- (e) If it is below freezing, it is also snowing.
- (f) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.
- (g) That it is below freezing is necessary and sufficient for it to be snowing.

3. Determine whether each of these conditional statements is true or false.

- (a) If $1 + 1 = 2$, then $2 + 2 = 5$.
- (b) If $1 + 1 = 3$, then $2 + 2 = 4$.

- (c) If $1 + 1 = 3$, then $2 + 2 = 5$.
- (d) If monkeys can fly, then $1 + 1 = 3$.

§1.3 Propositional Equivalences

4. Use truth tables to verify these equivalences.

- (a) $p \wedge T \equiv p$
- (b) $p \vee F \equiv p$
- (c) $p \wedge F \equiv F$
- (d) $p \vee T \equiv T$
- (e) $p \vee p \equiv p$
- (f) $p \wedge p \equiv p$

5. Use De Morgan's laws to find the negation of each of the following statements.

- (a) Kwame will take a job in industry or go to graduate school.
- (b) Yoshiko knows Java and calculus.
- (c) James is young and strong.
- (d) Rita will move to Oregon or Washington.

6. Determine whether $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$ is a tautology.

7. Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

8. Determine whether each of these compound propositions is satisfiable.

- (a) $(p \vee \neg q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q)$
- (b) $(p \rightarrow q) \wedge (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$
- (c) $(p \leftrightarrow q) \wedge (\neg p \leftrightarrow q)$

Wednesday — §1.4 Predicates & Quantifiers

9. Let $P(x)$ denote the statement “ $x \leq 4$.” What are these truth values?

- (a) $P(0)$
- (b) $P(4)$
- (c) $P(6)$

10. Let $P(x)$ be the statement “ x can speak Russian” and let $Q(x)$ be the statement “ x knows the computer language C++.” Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school.

- (a) There is a student at your school who can speak Russian and who knows C++.
- (b) There is a student at your school who can speak Russian but who doesn’t know C++.
- (c) Every student at your school either can speak Russian or knows C++.
- (d) No student at your school can speak Russian or knows C++.

11. Let $C(x)$ be the statement “ x has a cat,” let $D(x)$ be the statement “ x has a dog,” and let $F(x)$ be the statement “ x has a ferret.” Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

- (a) A student in your class has a cat, a dog, and a ferret.
- (b) All students in your class have a cat, a dog, or a ferret.
- (c) Some student in your class has a cat and a ferret, but not a dog.
- (d) No student in your class has a cat, a dog, and a ferret.
- (e) For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

12. For each of these statements find a domain for which the statement is true and a domain for which the statement is false.

- (a) Everyone is studying discrete mathematics.
- (b) Everyone is older than 21 years.
- (c) Every two people have the same mother.
- (d) No two different people have the same grandmother.

13. Express the negation of these propositions using quantifiers, and then express the negation in English.

- (a) Some drivers do not obey the speed limit.
- (b) All Swedish movies are serious.
- (c) No one can keep a secret.
- (d) There is someone in this class who does not have a good attitude.

Thursday — §1.5 Nested Quantifiers & start §1.6 Inference

§1.5 Nested Quantifiers

14. Let $Q(x, y)$ be the statement “ x has sent an e-mail message to y ,” where the domain for both x and y consists of all students in your class. Express each of these quantifications in English.

- (a) $\exists x \exists y Q(x, y)$
- (b) $\exists x \forall y Q(x, y)$
- (c) $\forall x \exists y Q(x, y)$
- (d) $\exists y \forall x Q(x, y)$
- (e) $\forall y \exists x Q(x, y)$
- (f) $\forall x \forall y Q(x, y)$

15. Let $L(x, y)$ be the statement “ x loves y ,” where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.

- (a) Everybody loves Jerry.
- (b) Everybody loves somebody.
- (c) There is somebody whom everybody loves.
- (d) Nobody loves everybody.
- (e) There is somebody whom Lydia does not love.
- (f) There is somebody whom no one loves.
- (g) There is exactly one person whom everybody loves.
- (h) There are exactly two people whom Lynn loves.
- (i) Everyone loves himself or herself.
- (j) There is someone who loves no one besides himself or herself.

16. Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

- (a) $\exists x \forall y (x + y = y)$
- (b) $\forall x \forall y (((x \geq 0) \wedge (y < 0)) \rightarrow (x - y > 0))$
- (c) $\exists x \exists y (((x \leq 0) \wedge (y \leq 0)) \wedge (x - y > 0))$
- (d) $\forall x \forall y ((xy = 0) \leftrightarrow ((x = 0) \vee (y = 0)))$

17. Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

- (a) $\neg \exists y \exists x P(x, y)$
- (b) $\neg \forall x \exists y P(x, y)$

- (c) $\neg \exists y (Q(y) \wedge \forall x \neg R(x, y))$
- (d) $\neg \exists y (\exists x R(x, y) \vee \forall x S(x, y))$
- (e) $\neg \exists y (\forall x \exists z T(x, y, z) \vee \exists x \forall z U(x, y, z))$

§1.6 Rules of Inference (start)

18. Find the argument form for the following argument and determine whether it is valid. Can we conclude that the conclusion is true if the premises are true?

If Socrates is human, then Socrates is mortal.
 Socrates is human.
 $\therefore$ Socrates is mortal.

19. What rule of inference is used in each of these arguments?

- (a) Alice is a mathematics major. Therefore, Alice is either a mathematics major or a computer science major.
- (b) Jerry is a mathematics major and a computer science major. Therefore, Jerry is a mathematics major.
- (c) If it is rainy, then the pool will be closed. It is rainy. Therefore, the pool is closed.
- (d) If it snows today, the university will close. The university is not closed today. Therefore, it did not snow today.
- (e) If I go swimming, then I will stay in the sun too long. If I stay in the sun too long, then I will sunburn. Therefore, if I go swimming, then I will sunburn.

Friday — §1.6 Rules of Inference (finish) & Fallacies

20. What rule of inference is used in each of these arguments?

- (a) Kangaroos live in Australia and are marsupials. Therefore, kangaroos are marsupials.
- (b) It is either hotter than 100 degrees today or the pollution is dangerous. It is less than 100 degrees outside today. Therefore, the pollution is dangerous.
- (c) Linda is an excellent swimmer. If Linda is an excellent swimmer, then she can work as a lifeguard. Therefore, Linda can work as a lifeguard.
- (d) Steve will work at a computer company this summer. Therefore, this summer Steve will work at a computer company or he will be a beach bum.
- (e) If I work all night on this homework, then I can answer all the exercises. If I answer all the exercises, I will understand the material. Therefore, if I work all night on this homework, then I will understand the material.

21. For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

- (a) “If I play hockey, then I am sore the next day.” “I use the whirlpool if I am sore.” “I did not use the whirlpool.”
- (b) “If I work, it is either sunny or partly sunny.” “I worked last Monday or I worked last Friday.” “It was not sunny on Tuesday.” “It was not partly sunny on Friday.”
- (c) “All insects have six legs.” “Dragonflies are insects.” “Spiders do not have six legs.” “Spiders eat dragonflies.”
- (d) “Every student has an Internet account.” “Homer does not have an Internet account.” “Maggie has an Internet account.”
- (e) “All foods that are healthy to eat do not taste good.” “Tofu is healthy to eat.” “You only eat what tastes good.” “You do not eat tofu.” “Cheeseburgers are not healthy to eat.”
- (f) “I am either dreaming or hallucinating.” “I am not dreaming.” “If I am hallucinating, I see elephants running down the road.”

22. For each of these arguments, explain which rules of inference are used for each step.

- (a) “Doug, a student in this class, knows how to write programs in JAVA. Everyone who knows how to write programs in JAVA can get a high-paying job. Therefore, someone in this class can get a high-paying job.”
- (b) “Somebody in this class enjoys whale watching. Every person who enjoys whale watching cares about ocean pollution. Therefore, there is a person in this class who cares about ocean pollution.”
- (c) “Each of the 93 students in this class owns a personal computer. Everyone who owns a personal computer can use a word processing program. Therefore, Zeke, a student in this class, can use a word processing program.”

- (d) “Everyone in New Jersey lives within 50 miles of the ocean. Someone in New Jersey has never seen the ocean. Therefore, someone who lives within 50 miles of the ocean has never seen the ocean.”

Saturday — §1.7 Introduction to Proofs & §1.8 Proof Methods

§1.7 Introduction to Proofs

- 23.** Use a direct proof to show that the sum of two odd integers is even.
- 24.** Use a proof by contradiction to prove that the sum of an irrational number and a rational number is irrational.
- 25.** Prove that if m and n are integers and mn is even, then m is even or n is even.
- 26.** Let $P(n)$ be the proposition “If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$.” Prove that $P(1)$ is true. What kind of proof did you use?
- 27.** Find a counterexample to the statement that every positive integer can be written as the sum of the squares of three integers.

§1.8 Proof Methods and Strategy

- 28.** Prove that $n^2 + 1 \geq 2n$ when n is a positive integer with $1 \leq n \leq 4$.
- 29.** Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a , b , and c are real numbers.
- 30.** Prove that there is a positive integer that equals the sum of the positive integers not exceeding it. Is your proof constructive or nonconstructive?
- 31.** Prove that either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?
- 32.** Prove that $\sqrt[3]{2}$ is irrational.

Sunday — §5.1 Mathematical Induction & Full Review

33. Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for the positive integer n .

- (a) What is the statement $P(1)$?
- (b) Show that $P(1)$ is true, completing the basis step of the proof.
- (c) What is the inductive hypothesis?
- (d) What do you need to prove in the inductive step?
- (e) Complete the inductive step.
- (f) Explain why these steps show that this formula is true whenever n is a positive integer.

34. Prove that $1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1}n^2 = (-1)^{n-1}\frac{n(n+1)}{2}$ whenever n is a positive integer.

35. Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1.

- (a) What is the statement $P(2)$?
- (b) Show that $P(2)$ is true, completing the basis step of the proof.
- (c) What is the inductive hypothesis?
- (d) What do you need to prove in the inductive step?
- (e) Complete the inductive step.
- (f) Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

36. Prove that $2^n > n^2$ if n is an integer greater than 4.

37. Prove that 2 divides $n^2 + n$ whenever n is a positive integer.

38. What is wrong with this “proof” that all horses are the same color?

Let $P(n)$ be the proposition that all the horses in a set of n horses are the same color.

Basis Step: Clearly, $P(1)$ is true.

Inductive Step: Assume that $P(k)$ is true, so that all the horses in any set of k horses are the same color. Consider any $k+1$ horses; number these as horses $1, 2, 3, \dots, k, k+1$. Now the first k of these horses all must have the same color, and the last k of these must also have the same color. Because the set of the first k horses and the set of the last k horses overlap, all $k+1$ must be the same color. This shows that $P(k+1)$ is true and finishes the proof by induction.